👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 4 Compression (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Week 3: Compression Handout + Compression Article")	Level 2 piece
Compression Handout (.doc)	Deck + notes A-C + quiz
Compression Article (scanned image)	Level 3 reading replaces it (our own text)
Class slides	Deck: 15 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 12 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Demo: squeeze a syringe or bike pump with a finger over the end to feel compression

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Compression Handout: L1 notes + quiz; L2 notes + quiz; same points
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **Compression Article**: The article is a scanned image (source unknown); our reading covers the same ideas.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Compression Test (Level 2)

Small Engines Overhaul · Unit 4 · goes with the class “Compression Handout” (lab points). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Compression Handout”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (lab points)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: Compression spec depends on the engines in Mr. Davis's shop; staff fill in the spec.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

💨 Compression Notes

Small Engines Overhaul · Unit 4 Compression · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Sealing

1. Match the part to where it seals:

Piston rings	head to block
Valves	plug hole
Head gasket	piston to cylinder wall
Spark plug washer	ports in the head/block

B. Low compression

2. Worn _____ let pressure leak past the piston.

3. A blown head _____ leaks between head and block.

4. Low compression makes the engine hard to _____.

Word bank: **gasket** **rings** **start**

C. The test

5. Number the steps:

_____ Install the gauge

_____ Record and compare to spec

_____ Remove the spark plug

_____ Pull the rope several times

6. Pressure is measured in _____.

Word bank: **PSI**

Level 2 version of the gen ed “Compression Handout + Compression Article.”

📖 Compression: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Compression

Squeezing the air-fuel mixture into a small space before the spark.



Piston rings

Metal rings around the piston that seal it against the cylinder wall.



Valve seat

The ring the valve presses against to seal the cylinder.



Head gasket

A thin seal between the cylinder head and the block.



Compression gauge

A gauge screwed into the spark plug hole to measure pressure in PSI.



PSI

Pounds per square inch, the unit for pressure.



Leak

When pressure escapes past worn rings, valves or a bad gasket.



Low compression

Not enough pressure; the engine is hard to start and weak.

📟 Compression Test

Small Engines Overhaul · Unit 4 · Level 2



Name: _____

Modified from the class “Compression Handout”.

🧺 Supplies

- ☐ 📟 **Compression gauge** with adapter
- ☐ 🔧 **Spark plug socket**
- ☐ ⚙️ **Practice engine**
- ☐ 🛢️ **Oil can** for wet test, L2/L3
- ☐ 📋 **Data sheet**

📟 Steps: check each box when you finish

1.  Disconnect the plug wire; remove the spark plug. ☐
2.  Thread the gauge in by hand. ☐
3.  Pull the rope 5 times; record PSI (dry test). ☐
4.  With adult supervision, add a teaspoon of oil to the cylinder. ☐
5.  Pull 5 times; record PSI (wet test). ☐
6.  Compare both to the spec. ☐
7.  Reinstall the plug; reconnect the wire. ☐

🚧 **Under construction:** Compression spec depends on the engines in Mr. Davis's shop; staff fill in the spec. *Check with the class teacher.*

📟 Compression Test · Prep Sheet

Small Engines Overhaul · Unit 4 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔧 **Mechanic**	Does the main engine job with the tools	
🧰 **Tool helper**	Gets tools from the board and returns them to staff	
📋 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Wipes the bench, sweeps, checks tools are back	

🧰 Tools and supplies: check when you have it

- ☐ 📟 **Compression gauge** with adapter
- ☐ ⚙️ **Practice engine**
- ☐ 📋 **Data sheet**
- ☐ 🔧 **Spark plug socket**
- ☐ 🛢️ **Oil can** for wet test, L2/L3

⚠️ Safety and manners focus

- 🥽 **Safety glasses** on the whole time.
- 👖 **Coveralls** and closed-toe shoes; tie back long hair.
- 🔧 Use tools **with adult supervision**. Right tool for the job.
- 🛢️ Oil in the cylinder can spray from the plug hole: **face away**.
- 🧼 **Clean up** and wash hands when you finish.

✅ Before you start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

📟 Compression Test · Reflection

Small Engines Overhaul · Unit 4 · Level 2

Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🥽 Wore safety glasses and used tools safely

☐ 🔧 Finished the Compression Test steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about engines?

1. _____
2. _____

Ideas: **The engine needs…** **I saw…** **Next time…**

✅ Compression Quiz

Small Engines Overhaul · Unit 4 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🗜️ Compression is…

A. Cooling the engine | B. Squeezing the air-fuel mix | C. Adding oil

2. 💥 A squeezed mix burns…

A. Weaker | B. Not at all | C. Hotter and with more push

3. ⭕ Piston rings seal…

A. The piston to the cylinder wall | B. The gas cap | C. The air filter

4. 🚪 A burned valve causes…

A. High compression | B. Better power | C. Low compression

5. 🧾 The head gasket seals…

A. Head to block | B. Tank to carb | C. Plug to wire

6. 📟 The compression gauge goes in the…

A. Oil fill | B. Spark plug hole | C. Muffler

7. 🔢 Compression is measured in…

A. Inches | B. PSI | C. Gallons

8. 📉 Low compression makes an engine…

A. Hard to start and weak | B. Faster | C. Quieter only

9. 🔓 A compression release makes…

A. More gas | B. More oil | C. Pulling the rope easier

10. 📘 Compare your reading to…

A. The manual's spec | B. A friend's guess | C. Nothing

💨 Compression Notes · ANSWER KEY

Small Engines Overhaul · Unit 4 Compression · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Sealing

1. Match the part to where it seals:

(Key: Piston rings → piston to cylinder wall · Valves → ports in the head/block · Head gasket → head to block · Spark plug washer → plug hole)

Piston rings	head to block
Valves	plug hole
Head gasket	piston to cylinder wall
Spark plug washer	ports in the head/block

B. Low compression

2. Worn **rings** let pressure leak past the piston.

3. A blown head **gasket** leaks between head and block.

4. Low compression makes the engine hard to **start**.

Word bank: **gasket** **rings** **start**

C. The test

5. Number the steps:

2 Install the gauge

4 Record and compare to spec

1 Remove the spark plug

3 Pull the rope several times

6. Pressure is measured in **PSI**.

Word bank: **PSI**

Level 2 version of the gen ed "Compression Handout + Compression Article."
🔑 Key written by our team from the class teacher's materials, please check.

✅ Compression Quiz · ANSWER KEY

Small Engines Overhaul · Unit 4 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🗜️ Compression is… Compression Handout

A. Cooling the engine | **B. Squeezing the air-fuel mix** | C. Adding oil

2. 💥 A squeezed mix burns… Compression Handout

A. Weaker | B. Not at all | **C. Hotter and with more push**

3. ⭕ Piston rings seal… Compression Handout

A. The piston to the cylinder wall | B. The gas cap | C. The air filter

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A. More gas | B. More oil | **C. Pulling the rope easier**

10. 📘 Compare your reading to… Notes

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Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "Compression Handout". Score: __ / 10. 🔑 Key written by our team from Mr. Davis's materials, please check.